

CariSurg MedTech Pathways 2026

Week 7 Draft Benchmark Comparison Table

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Programme: CariSurg MedTech Pathways 2026

GitHub Repository

<https://github.com/baldiea75-tech/carisurg-portfolio.git>

Comparing: Week 6 baseline models against the Week 7 complex model (random forest)

Split: identical to Week 6, 80/20 stratified on esi, random seed 42, reused rather than redrawn

Test set: 11,025 patients, of whom 16 are truly ESI 1 (the most critical level)

Full Comparison

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1	Recall (ESI 1)	Train Time	Inference (per patient)
Dummy baseline	0.375	0.204	0.204	0.204	0.375	0	0.002s	0.0003ms
Logistic Regression (Week 6)	0.667	0.582	0.463	0.492	0.661	0.25	2.7s	0.001ms
Decision Tree (Week 6)	0.556	0.265	0.245	0.216	0.463	0	0.7s	0.002ms
Random Forest (Week 7)	0.58	0.387	0.261	0.238	0.5	0	7.7s	0.017ms
Random Forest, balanced (Week 7)	0.478	0.395	0.492	0.369	0.481	0.25	7.5s	0.018ms
Random Forest, best trade-off found (Week 7)	0.47	0.392	0.549	0.367	0.476	0.562	2.7s	0.007ms

The best trade-off row comes from a wider search over tree depth and class weighting (`max_depth=8, class_weight='balanced'`). It more than doubles logistic regression's recall on ESI 1, but only by flagging 102 patients as ESI 1 in total to catch 9 true cases, a precision of 0.088 on that class, and by giving up nearly twenty points of overall accuracy. Columns are metrics. Rows are models, with the Week 6 baselines and the two Week 7 random forest variants clearly labelled. All values are rounded to 3 decimal places for accuracy, precision, recall and F1, and to whole seconds or milliseconds for timing, so this table can be read at a glance.

Core Four Metrics Only (Quick Reference)

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Logistic Regression (Week 6 baseline)	0.667	0.582	0.463	0.492
Random Forest (Week 7 complex model)	0.58	0.387	0.261	0.238

The complex model does not beat the Week 6 baseline on any of the four core metrics. Full per class detail, the recall on ESI 1 specifically and the reasoning behind the recommendation are in [docs/week7_cost_benefit_memo.md](#).